Transparency without price differentiation: credit-level evidence from the collapse of the first tokenized carbon pool

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Abstract

Carbon credits are meant to finance real climate benefits, yet credits sold as equivalent can differ sharply in what they deliver. A project may have gone ahead without carbon revenue; its claimed reductions may be hard to verify or may not last. Transparency is meant to address this problem: reveal information and let participants distinguish quality. We test that proposition in the Base Carbon Tonne (BCT) pool, a tokenized carbon market with a public transaction record. BCT placed 22 million tonnes of credits on a blockchain but issued one interchangeable token for every credit. Depositors received the same generic token; a holder could later choose an underlying credit to redeem. We reconstruct 1,187 deposits and 35,432 recorded withdrawals, link tokens to registry records, and assess their underlying credits using six observable features. BCT's collapse was widely blamed on forest-conservation credits (REDD+), but they made up only 4.2% of deposits. Old renewable-energy credits, many from projects likely to have been built anyway, made up 69.1%. A small set of accounts withdrew most credits with outside demand, leaving 9.3 million tonnes of largely low-additionality credits unredeemed. For credits represented in both BCT and a pool that restricted eligible deposits, redemption was much lower in the restricted pool, a descriptive pattern consistent with design affecting credit fate. The case shows why disclosure may not protect a uniform-price pool: participants can exploit quality differences that the pool price does not reflect. Carbon-market reforms and tokenized-asset designs may therefore need quality-differentiated prices, not disclosure alone.

Introduction

Carbon credits let a buyer claim that one tonne of its emissions has been balanced by one tonne kept out of the atmosphere elsewhere. That promise is valuable only if the credited climate benefit is real. Yet a credit's value cannot be read from the tonne printed on it. Did the project need carbon revenue to happen at all? Can its claimed reductions be measured and independently checked? If carbon is stored, how long will it remain out of the atmosphere? These are the differences that determine credit quality.

The most consequential of them is additionality: a credit is additional only when its revenue changes what would otherwise have happened. A wind farm that would have been built anyway can generate electricity and valid paperwork while delivering no extra climate benefit from the credit sale. This counterfactual is inherently difficult to observe. It helps explain why the voluntary carbon market (VCM), where companies buy credits outside mandatory cap-and-trade systems, has long struggled to distinguish strong credits from weak ones. Its annual transaction value peaked above \$2 billion in 2021–2022, yet fewer than 16% of credits from investigated projects represent genuine reductions[4,5].

Economics predicts a problem when markets cannot make that distinction. In Akerlof's "lemons problem", low-quality goods crowd out high-quality goods because buyers cannot tell them apart[1,2,3]. Carbon markets are often described in those terms. The proposed cure is transparency: make a credit's history and quality information visible, so participants can avoid poor credits and reward better ones.

Blockchain-based tokenization promised to put that cure into practice. It records transactions on a public ledger rather than in private, over-the-counter deals. The largest early experiment was Toucan Protocol's Base Carbon Tonne (BCT) pool. A holder could send a credit verified by Verra, the world's largest carbon-credit registry, to Toucan's bridge; the credit was represented on-chain and accepted into BCT. In return, the holder received an interchangeable BCT token. Every deposited credit earned the same token and every token traded at the same price, regardless of the underlying credit's quality. A later holder could return a BCT token and choose a particular underlying credit to withdraw. The pool therefore made quality visible, but did not price it. Its token fell from more than \$8 per tonne in late 2021 to \$0.71 by mid-2023 and effectively zero by 2026.

This design creates a question that conventional carbon markets cannot answer cleanly. If participants can inspect quality, will transparency alone prevent a pool of mixed credits from deteriorating? Conventional markets leave little public transaction data with which to follow each credit from entry to exit. BCT does not: its ledger records every deposit, withdrawal, and account. It is therefore a rare opportunity to test a transparency-based market design against its complete observable history, rather than infer its behaviour from prices or a sample of projects.

The existing explanation of BCT's collapse was also never checked against that history. Commentators and academics blamed low-quality forest-conservation credits, called REDD+ (a programme that pays landowners to avoid deforestation)[2,6,7,8,9]. That account fits extensive evidence of REDD+ crediting problems[4,10,11]. But it says nothing about whether such credits actually dominated this pool. More importantly, it leaves the design question open: would public information have protected the pool if its composition had been different?

We answer both questions by reconstructing BCT from the chain. We collect every deposit (1,187) and every recorded withdrawal (35,432), identify the 509 depositing and 28,897 withdrawing accounts, and link each credit token to its registry project and vintage. For the 161 tokens with individual assessments, we evaluate six features that bear on climate quality: whether the credit represents removal rather than avoided emissions, additionality, the reliability of monitoring, reporting and verification, how permanent the claimed benefit is, credit vintage, and the registry and methodology used. We then use the public transfer record to ask three linked questions: what entered the pool, what was selectively withdrawn, and whether identical credits had different outcomes in a screened sister pool. This combination lets us study not only whether low-quality credits existed, but how market architecture shaped their movement.

The results reverse the prevailing account. REDD+ was only 4.2% of BCT by deposited tonnage. Renewable-energy credits, mainly grid-connected hydropower from China, India, Turkey, and Brazil, were 69.1%; evidence indicates that many of their projects would have been built without carbon revenue. A small set of accounts withdrew most high-demand credits (1.55 million tonnes), while 9.3 million tonnes of low-value credits remain stranded. For 14 credit tokens that entered both BCT and a quality-screened sister pool, the same credits were fully withdrawn from BCT but only 28.5% withdrawn from the screened pool. This two-pool comparison is descriptive, not causal proof, but it is consistent with design rather than credit type shaping asset fate.

We call the resulting mechanism *design-enabled adverse selection*. It is not the classical lemons problem: quality-related information was publicly available, although participants need not have interpreted it alike. The key asymmetry was instead built into the pool. At entry, any eligible credit received the same generic token; at exit, a token holder could choose which credit to take. That is blind entry and selective exit. A single price gives no reward for bringing a better credit into the pool, while the option to choose at exit makes better credits worth extracting. The complete ledger lets us observe this mechanism at credit level, something opaque off-chain markets cannot provide.

The implications reach beyond one failed pool. Article 6.4 of the Paris Agreement is developing a new UN crediting mechanism, while more than $25 billion of other real-world assets are now being tokenized. Neither setting is automatically equivalent to BCT. But whenever a mechanism pools assets of different quality at one price, the design choice is testable and consequential: transparency may inform participants, yet quality-differentiated pricing and rules on both entry and exit may still be needed to protect the market.

Results

What entered BCT, and what the REDD+ narrative misses

We reconstructed all 1,187 deposits from BCT's public transaction ledger. They cover 168 projects in Verra's Verified Carbon Standard (VCS), a registry programme that records issued carbon credits, and about 22 million tonnes (Methods).

Renewable-energy credits dominated the pool (Fig. 1). They came from grid-connected hydro, wind, and solar projects, mainly in China and India and also in Turkey and Brazil, registered between 2008 and 2013; together they made up 69.1% of deposited tonnage. These projects used categories from the Clean Development Mechanism (CDM), the Kyoto Protocol crediting programme that ran from 2004 to 2020. The Integrity Council for the Voluntary Carbon Market (ICVCM), an industry governance body, has not approved these categories for its Core Carbon Principles (CCP) label. Programme-level assessments[12,13], a meta-analysis of wind-power CDM projects[4], and estimates for Indian wind offsets[5] all find little additionality: the projects would probably have been built without carbon revenue.

REDD+ credits, widely blamed for BCT's collapse, constituted just 4.2% of the pool, about one-sixteenth of the renewable share. Even when REDD+ is combined with the other nature-based categories, ARR (afforestation, reforestation, and revegetation) and IFM (improved forest management), the total reaches only 10%, versus 69.1% for renewables alone.

BCT's composition differed sharply from broad Verra issuance

BCT's renewable dominance might simply reflect what Verra issued more broadly. We therefore use published VCS issuance shares from MSCI[21] and Ecosystem Marketplace[24] as a broad supply reference: renewables account for approximately 37% of VCS issuance by volume (central estimate; sensitivity range 26–48%). This comparison measures a difference in composition relative to registry issuance; it does not identify why any depositor chose to bridge a credit.

Nearly nine in ten depositor accounts (89.0%) held portfolios in which renewables were the majority by tonnage. BCT's renewable share, 69.1%, was 1.87 times the 37% registry reference (a ratio of one would mean no compositional difference). Because one address can make many deposits, we treat addresses rather than transactions as the independent units; an account-level randomization test gives $p < 0.0001$ (Supplementary Methods). The difference remains under the highest 48% reference share (1.44 times) and when we check that it is not driven only by a small number of large accounts.

Conversely, REDD+ is under-represented at 0.2× the VCS base rate (BCT 4.2% vs. VCS ∼17%), directly contradicting the narrative that REDD+ credits overwhelmed the pool.

Where the skew arose. Toucan's bridge is the mechanism that represents a registry credit on-chain before it can enter a pool. Conditional on a credit being bridged, nearly all bridged tonnage entered BCT: 93.5% of token classes and 99.6% of tonnes (Supplementary Methods). The renewable skew therefore arose before, or at, the decision to bridge rather than from a later BCT admission screen. The ledger cannot distinguish depositor choice from other reasons why the bridged stream was renewable-heavy.

Price and incoming composition moved together

We next examine whether price and incoming composition moved together. We merged daily BCT prices in US dollars (1,493 observations, October 2021 to July 2026; Methods) with cumulative composition measures calculated from deposits ($n = 331$ overlapping days). This observational comparison cannot establish what drove the price decline.

Co-movement. Price and the cumulative tonnage-weighted mean quality score moved together (Pearson $r = 0.77$, $p < 10^{-66}$; Fig. 2a). For pool-level comparisons we also report its inverse, the Pool Quality Deficit (PQD $= 1 -$ score/100), for which a higher value means lower average quality; its correlation with price is -0.77. Because cumulative series share strong time structure, we also compare day-to-day changes. A one-percentage-point increase in the renewable share of deposits was associated with a \$0.018 lower BCT price on that specification ($p < 0.001$; $\beta = -1.8$ when the share is expressed from 0 to 1; Methods). Exploratory weekly tests of temporal precedence more often place a price change before a composition change. They are consistent with lower prices being followed by lower-quality deposits, but do not identify a feedback effect or causality (Supplementary Methods).

What left the pool. The 35,432 outbound transfer events show that credits were not withdrawn evenly. Industrial-gas credits were fully redeemed, as were 99.8% of REDD+, 93.0% of improved forest management (IFM), and 91.3% of afforestation, reforestation and revegetation (ARR) credits; only 3.7% of renewable-energy credits were redeemed. This pattern is consistent with selective choice at exit. Within the originally scored 161-token subset, redeemed credits had a tonnage-weighted mean composite score of 38.7 on a 0–100 scale, compared with 31.7 for deposited credits. Over the pool's lifetime, its net renewable share rose from 71% to 76%.

The order of withdrawals also followed differences among credit types. The types redeemed earliest were those with greater outside-market demand in our type-level comparison ($\rho = -0.74$, $n = 7$ types; Supplementary Methods). Before the May 2022 cryptocurrency crash, redeemed credits had a tonnage-weighted mean score of 42.0; afterwards it was 32.4. This timing is consistent with credits that retained outside value being withdrawn while a generic pool token still cost less, but the ledger alone does not reveal a redeemer's motive.

Depositors and redeemers are almost entirely distinct populations: only 1.4% of 28,897 redeemer accounts also appear among the 509 depositor accounts. The top 10 redeemers account for 85.0% of extracted tonnage. Their transactions occurred in rapid bursts (median gap 4.6 seconds), consistent with programmatic execution.

The ledger separates two facts that are often conflated. BCT's admission rule did not screen the credits already represented on the bridge, whereas withdrawals were strongly concentrated in particular types. The first fact describes a permissive architecture; the second describes selective exit. It does not establish that the same people, or even the same entities, drove both processes.

Account forensics: who extracted and what they took

We next trace the accounts behind withdrawals from the 161 originally scored token classes. The largest withdrawing addresses were concentrated in particular credit types, and most had not appeared among depositors.

Table 1 Top 5 redeemers by tonnage.

Account	Tonnes redeemed	Dominant type	Mean quality	Also depositor?
0x65a5...	651,334	Industrial gas	30.9	No
0x1b8e...	335,556	IFM	40.0	Yes
0xee4b...	195,051	REDD+	31.4	No
0x4b3e...	188,205	ARR	50.4	No
0x8556...	183,629	ARR	46.0	No

These five addresses withdrew 1.55 million tonnes: most redeemed industrial-gas credits, 37% of REDD+, and 44% of ARR. The largest address withdrew 651,334 tonnes of industrial-gas credits (42% of the total). It was a smart contract, an autonomous on-chain programme, that retired every credit in the same transaction in which it withdrew it (verified on-chain; Data availability). This pattern is consistent with an aggregator routing credits to final offset use. The other four were externally owned addresses, not identified individuals. Their post-withdrawal transfers and holdings are consistent with, but do not establish, value-seeking redemption. Under midpoint price assumptions, the potential spread is roughly \$7.4 million (\$10 million including the retirement contract; bounds \$0.3–\$22M; Methods). Fifteen of the twenty largest withdrawing addresses never deposited a credit, and each concentrated on one type.

Tracing each redeemed credit's immediate destination across the 20 largest extractors shows three pathways: cross-pool transfer to a quality-screened pool (NCT, Toucan's nature-restricted pool; 39.2%), immediate retirement (34.6%), and secondary-market sale (17.0%; Supplementary Methods).

Among the 399 addresses that both deposited and withdrew (a 1.4% overlap), a quality-swap analysis finds no systematic pattern of depositing weaker credits while withdrawing stronger ones (Supplementary Methods). The two sides of the observed pattern therefore involve largely different addresses. They need not involve wholly different entities. To examine that limit, we traced the first observable funder for the 20 largest addresses on each side. We found one shared funding source, direct credit transfers involving three addresses, and two addresses active on both sides. These observable links account for 26% of top-20 deposit tonnage and 9% of top-20 withdrawal tonnage (Supplementary Table 5); they do not resolve links hidden behind exchanges or intermediaries.

Quality atlas and quality gating

BCT's Pool Quality Deficit (PQD) is 0.679 in the 161-token individually scored subset. PQD is one minus the tonnage-weighted mean composite score divided by 100, so a higher value means lower average quality. This places BCT near the low-quality end of the spectrum in Fig. 3, which runs from 0.076 for direct air carbon capture and storage (DACCS) to 0.759 for grid-connected renewables. A permutation test finds no evidence that the deposited subset was lower quality than random draws from our defined eligible-token universe ($z = -0.64$, $p = 0.27$). It does not show why those credits became eligible or bridged. We translate the composite score into grades from B to AAA (Methods); BBB is our pre-specified first grade for credits with verified additionality. For a separate gate simulation using the full, 345-token deposit stream (with 184 scores imputed from matched project characteristics), a BBB floor reduces PQD from 0.689 to 0.506 but admits only 7.2% of deposited tonnage (Fig. 4).

Temporal dynamics and robustness

Deposit quality declined over BCT's operating life (Spearman $\rho = -0.24$, $n = 1{,}187$). In our decomposition, the decline is explained by the changing vintage mix: later deposits drew on progressively older vintages, the years in which the credited reductions occurred, and we find no additional decline within vintage (Supplementary Table 2). We use the May 2022 cryptocurrency crash as a dated descriptive landmark, not as an exogenous experiment on BCT. After it, deposit quality fell 3.4 points (Cohen's $d = 0.49$, a moderate effect), volume fell 98% (Fig. 2b), and the quality-decline slope tripled. The timing is consistent with price falling before deposit quality worsened, but does not establish a causal channel (Supplementary Methods).

Pool design and asset fate: a within-token cross-pool contrast

We identify 31 project–vintage token classes deposited into both the unscreened BCT pool and Toucan's Nature Carbon Tonne (NCT) pool, which restricted eligible deposits to nature-based credits. We then descriptively compare the 14 shared classes that were fully redeemed from BCT with their redemption from NCT. They represent 1.50 million tonnes. Only 28.5% of their NCT tonnage was redeemed. The mean token-level difference is +73.9 percentage points (95% CI [51, 93]; Fig. 5), with the same direction for every included class and credit type.

This is not an estimate of a causal pool-design effect. Inclusion in the 14-class sample depends on the BCT outcome, so the BCT-side 100% is partly built into the comparison and +73.9 points is an upper bound. The two pools also differed in exposure timing, liquidity, eligibility, and participants. Shared project–vintage classes hold the underlying project category and vintage constant, but do not remove those pool-level differences. We report the exact tests and sensitivity analysis in Supplementary Methods as a description of the observed pairs, not as evidence of independently assigned treatments.

What screening mechanically changes. We also recovered credit identity on-chain for five pools run by Toucan and C3, a second bridge protocol, and scored them with the same framework. Mean scores rise with screening strictness: about 29–31 in unscreened pools, 39–40 in nature-screened pools, and 77.9 in CHAR, a pool restricted to an allowlist of credit categories (Supplementary Table 4). This is not an independent test that screening causes better quality. The screens are defined by credit type, and our score is largely type-driven, so a screen mechanically changes the average score by changing which types it admits. Nature-based credits have almost identical within-type scores in screened and unscreened pools (differences ≤ 0.32 points; Supplementary Methods). The comparison illustrates the design choice, but the C3 pools have no external validation and it does not identify an effect within a type.

Redemption patterns are better described by credit type than by our composite grade

Credit type, rather than our quality framework, captures the main pattern of selective withdrawal. Using the pre-specified token-level prediction metric in Supplementary Table 3, a rule based on type accounts for withdrawal better than one based on quality grade. Within the 116 renewable tokens, neither vintage nor grade predicts withdrawal. The paper's core findings—composition relative to broad VCS issuance, withdrawal patterns, and the descriptive within-token comparison—therefore rest on transaction data and Verra credit-type labels, not on our scoring framework (Supplementary Table 6).

This leaves a large stranded inventory: 9.3 of 9.6 million deposited tonnes in the lowest grade band remain unwithdrawn. The individually scored analysis covers 15.2 million tonnes in total.

Discussion

In BCT, information sufficient to reconstruct provenance and many project attributes was publicly available, yet the pool became increasingly concentrated in types that our framework rates lower. We do not observe which participants inspected, understood, or trusted that information. The case therefore does not show that disclosure is useless. It shows that disclosure did not by itself make heterogeneous credits trade at quality-differentiated prices in this pool.

The relevant economic intuition resembles both the "lemons problem" and cheapest-to-deliver incentives. Akerlof's lemons model concerns hidden quality[1]; BCT is different because much of the information used to assess a credit was public. What we call design-enabled adverse selection arises instead from the contract's rules. Our ledger reconstruction provides descriptive evidence consistent with the lemons dynamics that Manshadi et al.[2] model theoretically, but it does not turn the two-pool comparison into causal identification.

The proposed mechanism requires a price difference that the pool does not reflect. If some underlying credits are valued more highly outside a one-price pool, holders have an incentive to target them at withdrawal, leaving less sought-after credits behind. In BCT, the withdrawal pattern and type-level outside-demand comparison are consistent with this prediction; the ledger does not observe each holder's valuation or motive. Traditional finance documents related incentives as cheapest-to-deliver pricing in the agency mortgage-backed-securities market[25,26,27,28]. BCT adds an observable credit-level record that traces the predicted sorting pattern, though not counterfactual prices or motives directly.

The misattribution of BCT's collapse to REDD+ credits[7,8,9] illustrates how structural quality failures can hide in plain sight. The CDM-era renewables that filled 69.1% of BCT had near-zero additionality: the projects were already economically competitive. Unlike REDD+ failures, which generate visible controversy over baselines (the no-project emission levels against which credits are measured) and deforestation, the additionality failure of renewable energy credits is structurally invisible: the dams and wind farms exist, the electricity is real. Quality assurance frameworks that focus on high-profile failure modes may miss the categories that dominate pool composition.

More than nine million tonnes of nominally valid offsets remain unredeemed (97.6% non-redemption within the scored subset). Under assumed additionality ranges, social-cost-of-carbon values, and a registry-average counterfactual composition, a Monte Carlo calculation puts the median climate-value gap at \$146M (90% simulation interval \$68M–\$252M; parameter uncertainty only). This is a sensitivity-based welfare estimate, not an observed loss. A separate price-assumption calculation gives a potential private spread of about \$10 million, including the retirement contract; it is not observed profit and should not be added to the welfare estimate. The largest withdrawing address retired credits rather than taking a trading profit. The figures respectively describe the pool's composition, a possible spread, and the circulation outcome.

The mechanism may extend beyond carbon, but that prediction requires further tests. As a descriptive comparison, we examine six uniform-priced NFT vaults, pools of unique digital collectibles on NFTX. Deposited items sit below their collection's median sale-value percentile in all six vaults (median percentiles 0.25–0.40, where 0.5 indicates no difference; Supplementary Methods). None shows a corresponding exit pattern. These data show entry-side composition differences in another setting; they do not replicate BCT's dual-margin mechanism or establish a common cause.

These findings apply most directly to voluntary pools and registry-based credit bundles in which heterogeneous units clear at one price[14,15]. Article 6.4 is a registry-based crediting mechanism, not a uniform-priced pool. It therefore inherits a design principle rather than a direct prediction: price quality, or expect selection wherever quality remains unpriced. A gate alone would not have been enough. Even a minimal quality floor (composite score $\geq$29) would have excluded 58% of BCT's tonnage.

The case motivates three design hypotheses for pooled crediting mechanisms: differentiate prices by quality rather than set one price for all units; consider rules at both entry and exit; and review quality floors as composition changes. These are hypotheses for implementation and testing, not effects identified by BCT alone. They are consistent with the ICVCM Core Carbon Principles framework[16] and relevant to emerging quality-differentiation systems, including Singapore's use of independent ratings in eligibility assessments[17] and blockchain credibility indices[18]. We provide an open-source reference implementation—a registry and gate on-chain plus an off-chain monitor—whose gate check requires about 19k gas[19] (Supplementary Methods). It is a prototype, not evidence that deployment will prevent market failure.

The same data illustrate a prospective diagnostic rather than validate an early-warning rule. Our cumulative Lemons Index is the running, deposit-weighted PQD on a 0–1 scale, where higher is worse. It was 0.71 at launch. Using the illustrative 0.50 threshold, it was already above the threshold while the price remained near its

$5.66 launch level, roughly seven months before the price fell below half that value. A framework-free alternative, cumulative renewable share, also remained above its threshold from the first week (Supplementary Methods). The signal is therefore BCT's initial composition, not a later trend. It motivates a practical protocol to test prospectively: audit composition at launch, apply quality rules to deposits and redemptions, and re-audit as the pool evolves.

Several limitations warrant note. Our quality scores are author-derived (commercial ratings cover only a handful of BCT projects), but the core findings rely on transaction data and Verra credit-type classifications, not on the scoring framework. The framework itself is calibrated against CCP labels ($d = 1.8$; Fig. 6), commercial ratings ($\rho = +0.901$, $n = 27$; non-blind, scored with agency ratings in view; Supplementary Fig. 1), and a large-language-model panel (Fleiss' $\kappa = 0.6$, moderate agreement; Supplementary Methods). Both cross-pool comparisons are descriptive, not causal identifications. The within-token contrast (+73.9 pp gap, $\Gamma = 3.05$ bound) and the pool-level difference-in-differences (comparing changes across the two pools) rest on only two pools; their statistics quantify per-pair variability, not the pool-level confounds (exposure timing, liquidity, eligibility) that remain collinear with the screening label (Methods). The dominant renewables are structurally excluded from this nature-based comparison. Finally, near-complete bridge-to-pool pass-through locates the renewable skew before or at the bridging decision, but the ledger cannot distinguish strategic depositor choice from other determinants of that stream.

The policy implication is falsifiable. The mechanism predicts vulnerability where heterogeneous credits are voluntarily pooled at one price and holders can target particular units at exit. Transparency reforms under the ICVCM and Article 6.4 may improve information, but BCT indicates that information alone is not necessarily sufficient under those conditions. Article 6.4 is not currently a uniform-price pool. If it, or another mechanism, were to pool heterogeneous credits at one price, the prediction can be tested against a design with quality-differentiated prices or redemption rules.

The tokenized real-world asset sector now exceeds $25 billion and includes pooled designs for commodities, bonds, and receivables. BCT is a documented case in which public information did not prevent quality sorting in a uniform-price carbon pool. Its lesson is conditional but practical: tokenization does not remove the need to decide how quality affects entry, price, and exit.

Methods

Full methods are provided in Supplementary Methods; this section summarizes the essentials.

Data. We use the public Polygon ledger because it permits a census rather than a sample of BCT transactions. Direct node queries recover all 1,187 BCT deposits from October 2021 to late 2022 (168 Verra VCS projects, 345 TCO2 tokens—Toucan's project–vintage credit tokens—and approximately 22 million tonnes), and all 708 NCT deposits. For each, we record transaction, depositing address, token, project, vintage, and tonnage. We link projects to Verra registry records for methodology category, country, and CCP status. We define a redemption as an ERC-20 transfer from the pool to an address, yielding 35,432 events for the 161 originally scored tokens. Daily BCT prices (1,493 observations, October 2021 to July 2026) come from DeFi Llama, a cross-exchange price aggregator, through the committed `fetch_bct_prices.py`. The ledger reveals transfers and addresses, not the off-chain identity or motive of their users.

Quality scoring. We develop a composite score because no commercial rater covers enough BCT projects to characterize the pool. The score is author-derived and is not the basis of the paper's composition or transfer findings. It combines six observable dimensions that the Oxford Offsetting Principles[20] and ICVCM CCP Assessment Framework[16] identify as central to credit integrity: removal type (weight 0.250), additionality (0.200), monitoring, reporting, and verification (MRV; 0.200), permanence (0.175), vintage (0.100), and registry/methodology (0.075). Grades from B to AAA are score bands, subject to seven disqualifiers. The Pool Quality Deficit, PQD $= 1 - \bar{C}/100$, is the tonnage-weighted inverse of the mean score; its running value is the Lemons Index used as a diagnostic. In 10,000 random reweightings that preserve a total weight of one, 93.7% of grades remain unchanged.

Composition and cross-pool comparisons. To describe BCT's composition relative to broad registry supply, we calculate $S = f_{\mathrm{BCT}}/f_{\mathrm{VCS}}$, using a 37% VCS renewable reference and a 26–48% sensitivity range. Because addresses can deposit repeatedly, inference is clustered by address. This is a comparison of composition, not a test of depositor intent. For the cross-pool analysis, we identify shared project–vintage token classes and report their redemption difference with exact tests and a percentile bootstrap. Inclusion in the 14 fully redeemed BCT classes depends on the BCT outcome, so the comparison is descriptive and its gap is an upper bound; a Rosenbaum-style Γ is reported only as a sensitivity calculation. A deposit-level difference-in-differences and five-pool comparison are likewise descriptive.

Price, cross-domain, and welfare analyses. We use first differences and Newey–West errors to describe short-run co-movement of price and composition while reducing shared trends ($n = 330$ daily changes); the weekly temporal-precedence tests ($n = 55$) are exploratory and not causal. For a limited cross-domain comparison, the NFTX pipeline gives each deposited or redeemed collectible the percentile of a nearby sale within a window designed to limit collection-wide price drift. It tests entry and exit patterns separately and excludes router accounts that batch transactions. Finally, a 10,000-iteration Monte Carlo calculation propagates uncertainty in literature-based additionality ranges[4,12,22,23], the social cost of carbon (\$50–\$200/tCO_2), and a registry-composition counterfactual. It is an illustrative welfare sensitivity analysis, not an observed monetary loss.

Inference. We control the false-discovery rate with the Benjamini–Hochberg procedure at $\alpha = 0.05$ across ten pre-specified primary tests: the address-clustered composition difference; full-sample and pre-shock temporal correlations; the within-token sign test; a descriptive pool-level difference; two temporal-precedence tests; the within-pool permutation; depositor concentration; and the token-level redemption difference ($n = 161$). A 10,000-iteration address-clustered bootstrap covers deposit-level statistics. Statistical significance quantifies compatibility with the stated null models; it does not remove the design and data limitations described above.

Data availability

All data, scoring rubrics, analysis scripts, and smart contract source code are available at https://github.com/Adeline117/carbon-neutrality (version tag `v1.0-erl-submission` pins the state cited here) under an MIT license. Machine-readable rubrics are in JSON format. The 318-credit methodology batch dataset with per-dimension scores is included. All 345/345 BCT tokens are scored (161 original + 184 imputed). On-chain deposit data for both BCT (1,187 transactions) and NCT (708 transactions) are provided with transaction hashes enabling independent verification. The externally-owned-account versus smart-contract status of the top redeemers was checked with `eth_getCode` against a public Polygon node; results are recorded in `redeemer_contract_check.json`. The entity-level independence audit (first-funder resolution for the top 20 accounts per margin) is reproducible via `entity_funding_analysis.py`, with per-wallet results in `entity_funding_analysis.json`. LLM panel outputs for all models across 29 credits are included. Analysis scripts are pure Python with NumPy/SciPy as sole dependencies; a single verification script (`tools/verify_headline_numbers.py`) re-asserts every headline number against the cached analysis outputs.

Code availability

All analysis code is available at https://github.com/Adeline117/carbon-neutrality under an MIT license.

Author contributions

A.W. designed the study, collected and analyzed all data, developed the quality scoring framework, performed all statistical analyses, and wrote the manuscript. W.C. supervised the research and provided critical revisions.

Competing interests

The authors declare no competing interests.

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References

[1] Akerlof, G. A. The market for "lemons": quality uncertainty and the market mechanism. *Q. J. Econ.* **84**, 488–500 (1970).

[2] Manshadi, V. H. et al. Offsetting carbon with lemons: adverse selection and certification in the voluntary carbon market. SSRN Working Paper (2025).

[3] Battocletti, V. et al. The voluntary carbon market: market failures and policy implications. *Colo. Law Rev.* **95**, 519–572 (2024).

[4] Probst, B. et al. Systematic assessment of the achieved emission reductions of carbon crediting projects. *Nat. Commun.* **15**, 9562 (2024).

[5] Calel, R. et al. Do carbon offsets offset carbon? *Am. Econ. J. Appl. Econ.* **17**, 1–40 (2025).

[6] Berg, F. et al. The market for voluntary carbon offsets. SSRN Working Paper (2025).

[7] Bassi, E. et al. Blockchain-based voluntary carbon market: strategic insights into network structure. *Front. Blockchain* **8**, 1603695 (2025).

[8] Ballesteros-Rodriguez, A., De-Lucio, J. & Sicilia, M.-A. Tokenized carbon credits in voluntary carbon markets: the case of KlimaDAO. *Front. Blockchain* **7**, 1474540 (2024).

[9] Jirasek, M. KlimaDAO: a crypto answer to carbon markets. *J. Organ. Des.* **12**, 271–283 (2023).

[10] Trencher, G. et al. Demand for low-quality offsets by major companies undermines climate integrity of the voluntary carbon market. *Nat. Commun.* **15**, 6863 (2024).

[11] West, T. A. P. et al. Action needed to make carbon offsets from forest conservation work for climate change mitigation. *Science* **381**, 873–877 (2023).

[12] Cames, M. et al. How additional is the Clean Development Mechanism? Oko-Institut Report (2016).

[13] Schneider, L. Assessing the additionality of CDM projects: practical experiences and lessons learned. *Clim. Policy* **9**, 242–254 (2009).

[14] World Bank. State and Trends of Carbon Pricing 2025. World Bank Group (2025).

[15] Garcia, A. & Sanford, L. On the potential for strategic behavior in jurisdictional REDD+. *Proc. Natl Acad. Sci. USA* **123**, e2531612123 (2026).

[16] Integrity Council for the Voluntary Carbon Market. Core Carbon Principles, Assessment Framework, and Assessment Procedure. ICVCM (2023).

[17] Singapore National Environment Agency. Carbon rating panel: appointment of BeZero, Calyx Global, and Sylvera. NEA Regulatory Notice (2025).

[18] Gao, H. O., Liu, X. et al. A blockchain-based carbon registry platform for credible climate action in transportation. *npj Clim. Action* **5**, 23 (2026).

[19] Reference implementation: composable quality-rating registry and gated pools (Solidity/Foundry). https://github.com/Adeline117/carbon-neutrality (contracts/), MIT license.

[20] Allen, M. et al. The Oxford Principles for Net Zero Aligned Carbon Offsetting. University of Oxford (2020).

[21] MSCI. State of integrity in the global carbon-credit market. MSCI ESG Research (2024).

[22] Badgley, G., Freeman, J., Hamman, J. J., Haya, B., Trugman, A. T., Anderegg, W. R. L. & Cullenward, D. Systematic over-crediting in California's forest carbon offsets program. *Glob. Change Biol.* **28**, 1433–1445 (2022).

[23] Haya, B. K. et al. Comprehensive review of carbon quantification by improved forest management offset protocols. *Front. For. Glob. Change* **6**, 958879 (2023).

[24] Ecosystem Marketplace. State of the Voluntary Carbon Market 2023. Forest Trends (2023).

[25] Vickery, J. I. & Wright, J. TBA trading and liquidity in the agency MBS market. *FRBNY Econ. Policy Rev.* **19**, 1–18 (2013).

[26] Fusari, N., Li, W., Liu, H. & Song, Z. Asset pricing with cohort-based trading in MBS markets. *J. Finance* **77**, 3249–3287 (2022).

[27] An, Y., Li, W. & Song, Z. TBA trading and security issuance in the agency MBS market. *Real Estate Econ.* **53**, 607–642 (2025).

[28] Huh, Y. & Kim, Y. S. Cheapest-to-deliver pricing, optimal MBS securitization, and welfare implications. *J. Financ. Econ.* **150**, 68–93 (2023).

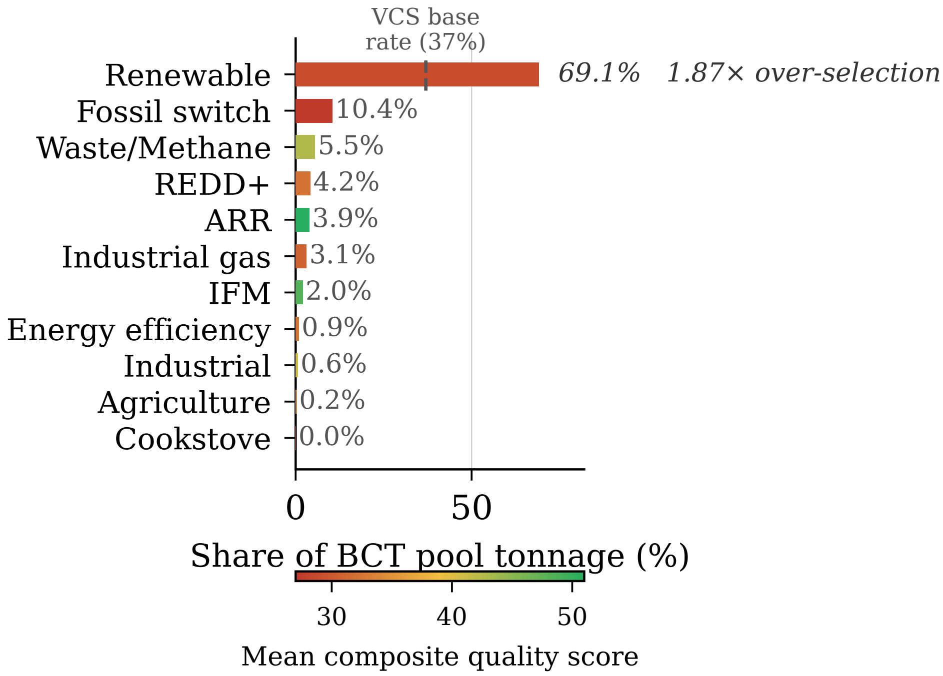

Fig. 1 BCT pool composition by methodology category, weighted by deposited tonnage. Renewable energy credits constitute 69.1% of total deposited tonnage; REDD+ constitutes just 4.2%.

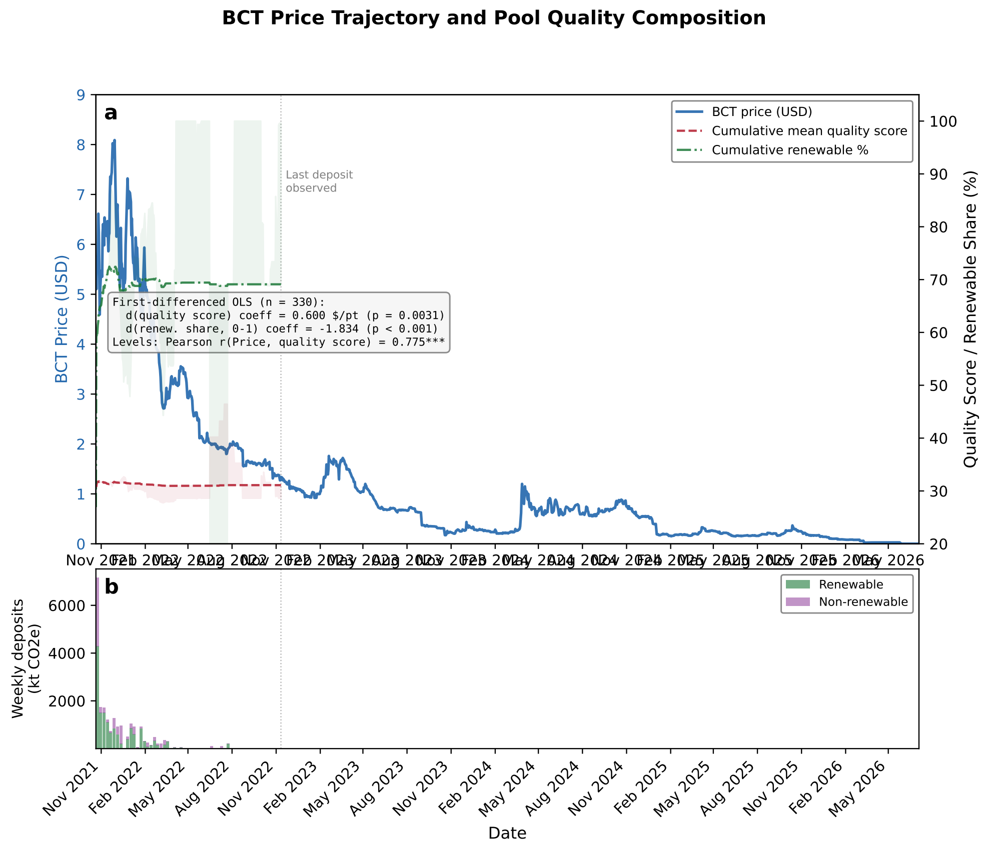

Fig. 2 BCT price trajectory and pool quality composition. (**a**) Daily BCT price against cumulative Pool Quality Deficit and cumulative renewable share; inset reports the first-differenced OLS (n = 330) and the levels correlation (Pearson $r = 0.774$). (**b**) Weekly deposit volume split into renewable and non-renewable tonnage.

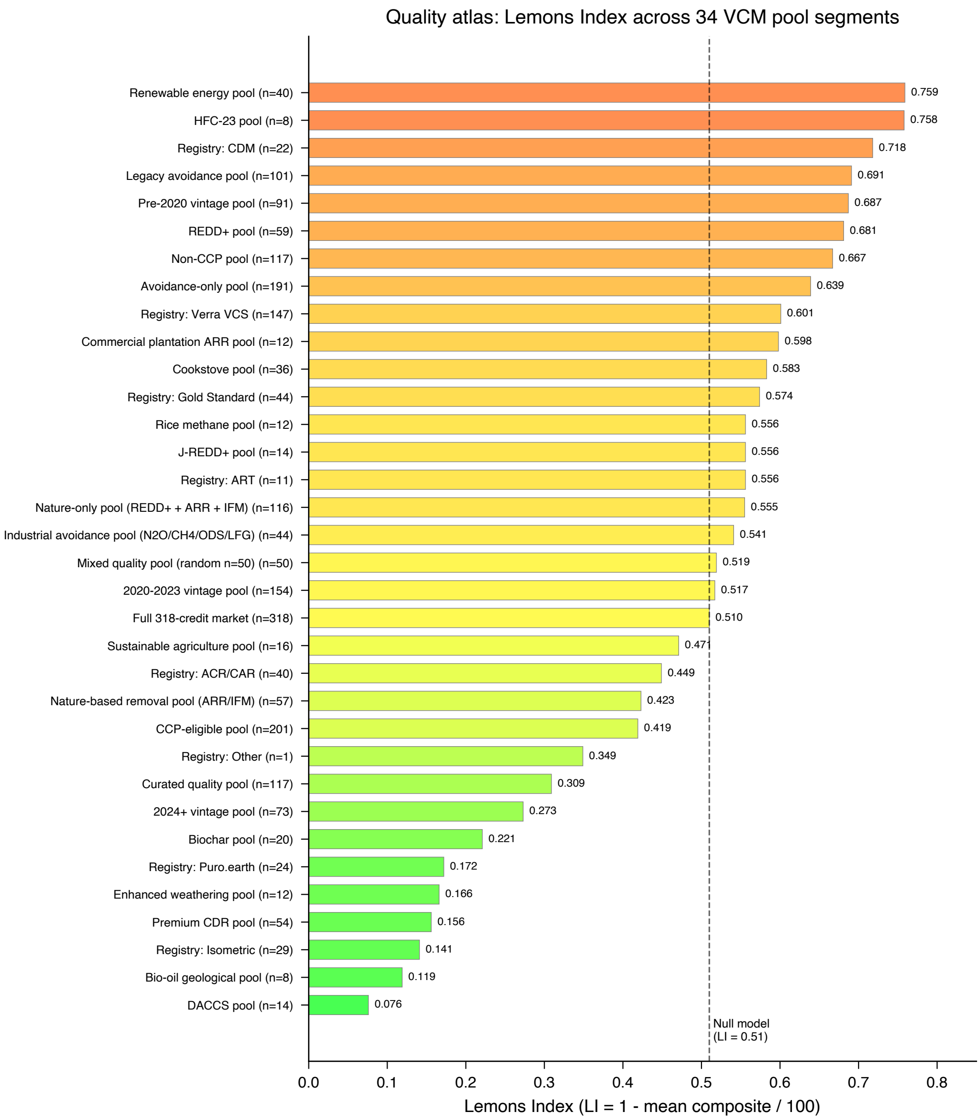

Fig. 3 Quality atlas: 34-segment PQD spectrum with null model baseline. PQD ranges from 0.076 (DACCS) to 0.759 (grid-connected renewables).

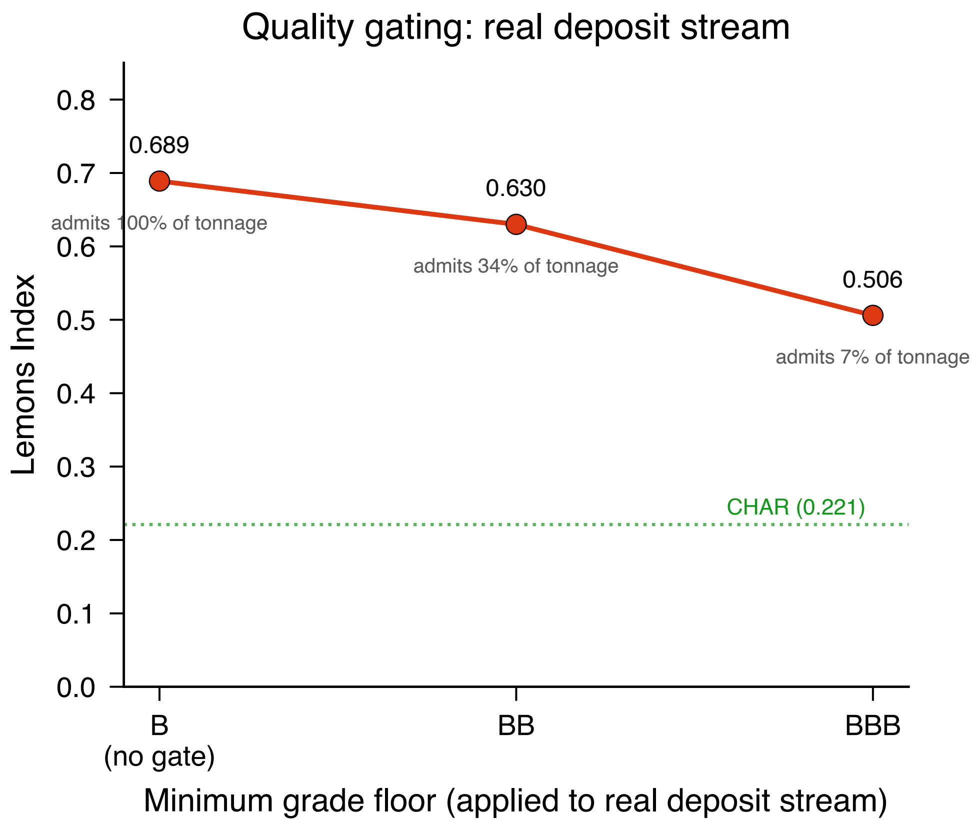

Fig. 4 Quality gating applied to the real BCT deposit stream: gated Lemons Index and admitted tonnage share under grade floors (B = no gate, BB, BBB). A BBB floor cuts the deposit-weighted Lemons Index (PQD) from 0.689 to 0.506 but admits only 7.2% of deposited tonnage; the CHAR category-allowlist pool (0.221) is shown for reference.

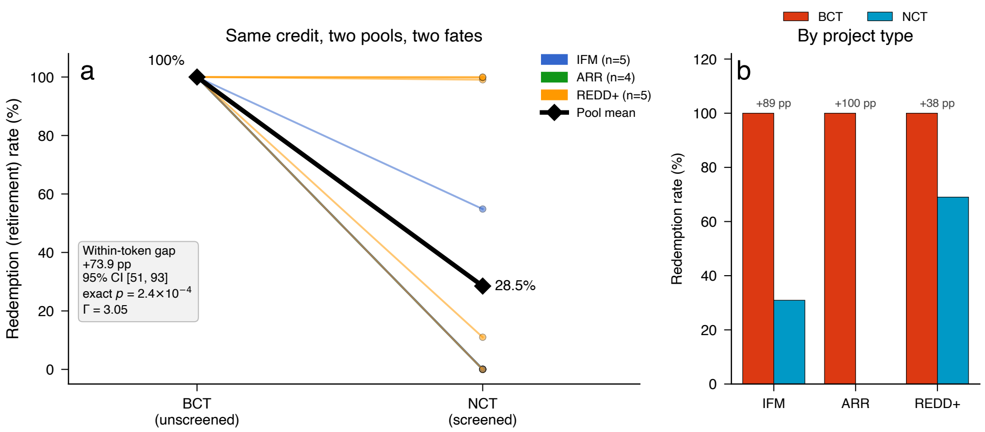

Fig. 5 Within-token cross-pool contrast: redemption of identical credits under two pool designs (descriptive; a two-pool comparison, not a causal identification). (**a**) Per-token slope plot of redemption (retirement) rate for the 14 carbon credit tokens deposited into *both* the unscreened BCT pool and the quality-screened NCT pool, colored by project type; the bold black line traces the pool means (100.0% redeemed from the unscreened pool versus 28.5% from the screened pool). Across the 14 pairs (the unit of inference) the mean within-token redemption gap is +73.9 percentage points (token-level bootstrap 95% CI [51, 93]); the direction is unanimous, so exact sign, permutation, and Wilcoxon tests all return $p = 2.4 \times 10^{-4}$ whether or not the one near-tied credit is included. (**b**) Redemption rate by project type (bars: tonnage-weighted rates; annotated gaps: per-token mean differences): the gap is positive within every type (IFM +89, ARR +100, REDD+ +38 pp per-token). The result is robust to unobserved confounding up to a Rosenbaum bound of $\Gamma = 3.05$. Depositor-level temporal, base-rate, and difference-in-differences robustness analyses are reported in Supplementary Methods.

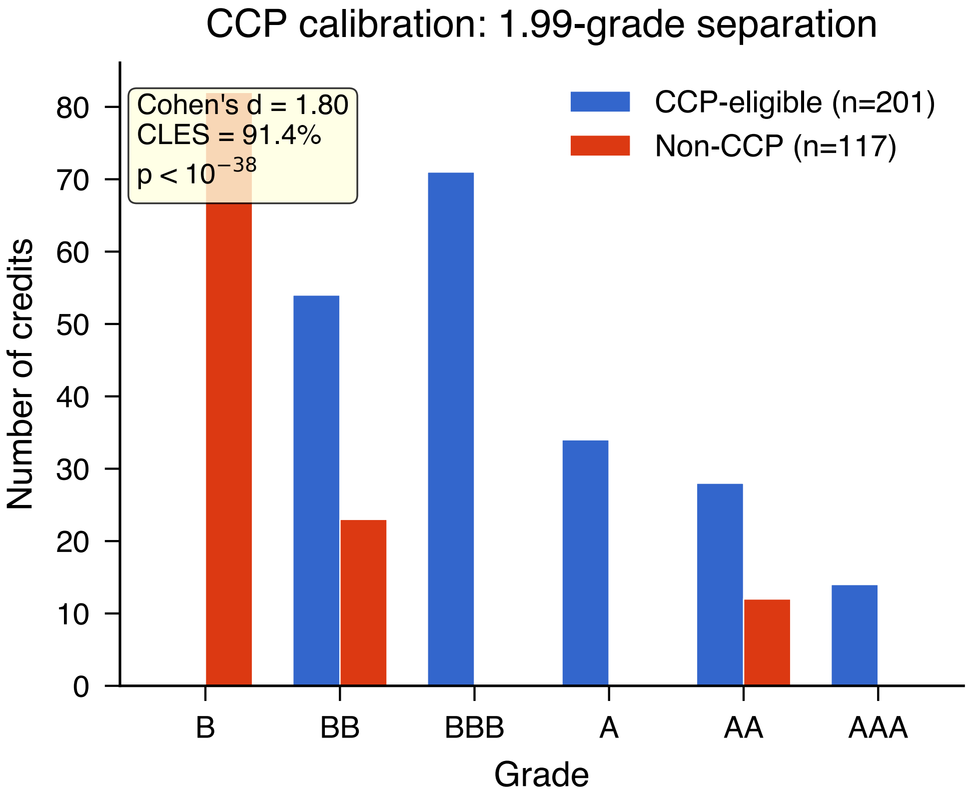

Fig. 6 CCP calibration: quality score distributions for CCP-eligible versus non-CCP credits. Cohen's d = 1.8 (95% CI [1.5, 2.16]). In-figure statistics: the title's 1.99 grades is the mean grade-band separation between the two groups; CLES is the common-language effect size (probability a random CCP-eligible credit outscores a random non-CCP credit, 91.4%).